

SECTION B

7 (short)

(a) An isotropic, linear elastic material with Young's modulus E and Poisson's ratio ν is subjected to tensile stresses σ_1 , σ_2 and σ_3 along its principal axes. Show that the resulting strains ε_1 , ε_2 and ε_3 are given by

$$\begin{aligned}\varepsilon_1 &= \frac{1}{E}(\sigma_1 - \nu\sigma_2 - \nu\sigma_3) \\ \varepsilon_2 &= \frac{1}{E}(-\nu\sigma_1 + \sigma_2 - \nu\sigma_3) \\ \varepsilon_3 &= \frac{1}{E}(-\nu\sigma_1 - \nu\sigma_2 + \sigma_3)\end{aligned}\tag{4}$$

(b) A cube of the material in part (a) is compressed along the 1 direction but is constrained in both transverse directions. Show that the “effective modulus” (σ_1/ε_1) relating the stress σ_1 and strain ε_1 along the 1 direction is given by

$$\frac{\sigma_1}{\varepsilon_1} = \frac{E(1-\nu)}{(1+\nu)(1-2\nu)}\tag{6}$$